

Numericals

$$\text{① EAT} = (\text{hit ratio} \times \text{memory access time}) + \left(\frac{\text{miss ratio}}{2} + 2 \times \text{memory access time} \right)$$

$$= (0.80 \times 10 \text{ nano sec}) + (0.20 + 2 \times 10) = 0.80 \times 10 + 0.20 \times 20$$

$$= 8 + 4 = 12 \text{ nano sec}$$

② RAM Size = 32 MB
Location Size = 1B
No of entries?

$$\frac{32 \text{ MB}}{1 \text{ B}} = \frac{32 \times 10^6}{2^5 \cdot 2^{20}} = 2^{25} \text{ entries in RAM.}$$

③ Logical Address = 26 bits, Physical address = 16 bits
No of Pages? No of frames? Page Size = 1KB = 2^{10}

No of Page = 2^{16}
No of frame = 2^6

16	10	Logical
6	10	Physical

④ No of Pages = 64, Page Size = 1024, No of frames = 32
Logical address space? Physical address space?

(i) Logical = $64 \times 1024 = 2^6 \cdot 2^{10} = 2^{16}$
(ii) Physical = $32 \times 1024 = 2^5 \cdot 2^{10} = 2^{15}$

⑤ Page Size = 1024 = 2^{10}
Length of address Register = 18
(a) Size of RAM?

$2^8 \cdot 2^{10}$
↓
256 KB

$$\text{frames} = \frac{256 \text{ KB}}{1 \text{ KB}} = 256 \text{ Frames}$$